

Remote Patient Monitoring System

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Public technical summary

A desktop application designed to help clinicians review real-time physiological signals, patient records, and alerts through a structured, extensible software architecture.

Problem

Clinical monitoring systems require a coherent view of live vital signs, historical information, and abnormal events without losing traceability across patient records.

System architecture

Java Swing client $\longleftrightarrow$ REST API $\longleftrightarrow$ PostgreSQL database

Implemented functionality

- Multi-patient monitoring interface with live ECG and respiration waveform visualisation.
- Patient onboarding, persistent clinical records, and threshold-based visual and audio alerts.
- Event acknowledgement and stored alert causes to support review and traceability.
- Digital-twin visualisation module as an extensible interface for representing physiological state.

System map

Presentation layer	Java Swing desktop interface for multi-patient views and live waveform display
Application layer	REST services coordinating clinician, patient, vital-sign, and alert workflows
Data layer	PostgreSQL persistence for structured patient data, historical records, and alert events
Quality controls	JUnit test suite and Gradle build workflow

Project outcome

The completed prototype provided a coherent clinical workflow from patient onboarding to live monitoring, alert acknowledgement, and review of stored health information. It demonstrated how modular software design can support a future clinical monitoring platform without coupling the interface to one specific hardware device.

Technical tools

Java Swing REST APIs PostgreSQL Gradle JUnit digital health software This public summary intentionally excludes credentials, database endpoints, and other security-sensitive implementation details.